GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for ESD and BDV Valves-BBOU/ELEP/AKAB**
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Specification for ESD and BDV Valves-BBOU/ELEP/AKAB

R02	19.10.22	Re-Issued for Review	P. Roy	P. Olukah	A. Duru	
R01	05.07.22	Issued for Review	P. Roy	P. Olukah	W. Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	05.07.22	Issued for Review
R02	19.10.22	Re-Issued for Review after incorporating TCE comments

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-038

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with ELEP & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS:

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development of a Gas Processing and Methanol Complex comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (ELEP), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with ELEP & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

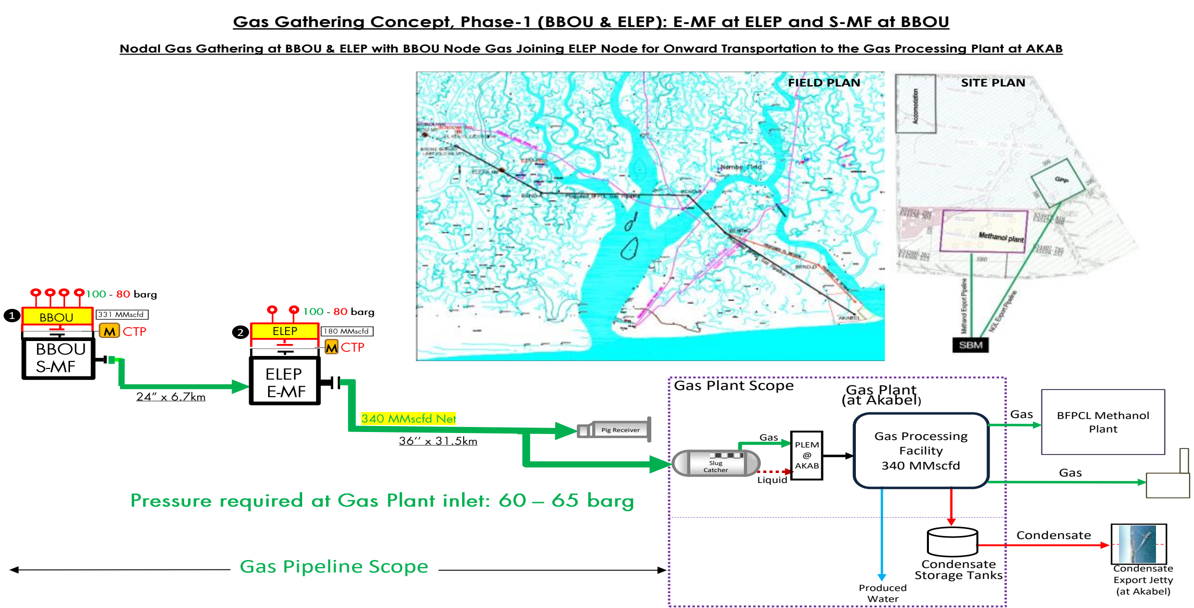

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, Inspection, testing, certification, delivery for ESD/BDV valves for BBOU and ELEP.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or sAupplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. General

3.1. The vendor shall study the requirement of SDV and BDV carefully and suggest the best possible technology for the application. The vendor shall be responsible for verification of the selection of valve size, type and material for each application in accordance with the information furnished.

3.2. Valve shall be suitable for Design condition.

3.3. Field instrumentation shall have a minimum ingress protection of IP65 in accordance with IEC 60529. All instruments shall be intrinsically safe or explosion proof and shall be certified to specific hazardous area class by statutory body such as ATEX, PTB.

3.4. Materials of Construction: Materials of construction for instrument equipment shall be consistent with corrosion resistance and the pressure and temperature conditions of the process fluids involved. Generally, the parts in contact with the process fluid shall be of a material equal to or better than that used for the related vessel or line.

3.5. Pressure & temperature rating

Pressure and temperature rating of instruments and components shall conform to the design rating of the process system to which they are connected. Instruments which can be exposed to vacuum shall have under-range protection to full vacuum.

3.6. Hazardous area requirement: Classification of Hazardous Areas shall be Zone1, IIA/IIB, T4. All instrumentation located or partially located in the hazardous area shall be suitable for classified hazardous area. All instruments and Junction Boxes shall be certified as Increased Safety (Exe). Power supply shall be switched off before opening the Instrument or Junction Box. ATEX/PTB approved certification shall be supplied for all electrical / electronic equipment in the hazardous area.

3.7. Accessibility: All instruments shall be accessible from grade or platform for measurement. In case of gas service where instruments are mounted above the taps, the installation of platform shall be considered, if accessibility is not available.

3.8. Vendor shall follow the international codes and standards and local regulations

3.9. Vendor shall offer inspection and testing as per the contractual requirement.

3.10. Vendor shall provide all related documentations like installation diagram etc.

4. Technical Specification for SDV and BDV Valves

4.1. ESD and BDV valves

4.1.1. Actuated valve type, materials and connections shall be in accordance with the pipe specification. All ESD and BDV valves shall be designed as per API 6D.

4.1.2. All ESD/BDV valve shall be SIL-2/3 certified. Final SIL level shall be decided during detail engineering after SIL study.

4.1.3. The ESD / BDV valve body size shall be same as line size. Ball valve shall be used.

4.1.4. Valve design shall be Trunnion mounted ball valve, Single-body/split body with built-in double block and bleed mechanism. Ball valve shall be suitable for bi-directional shut-off, unless otherwise specified.

4.1.5. Valve Body and Bonnet shall be meet the piping material specification and shall be generally carbon steel A105. Valve shall have 316 SS ball and trim, shall meet the piping material specification.

4.1.6. All valves shall have built-in fire-safe design.

4.1.7. All valves shall be soft seated with PTFE/PEEK material and shall be tight shut-off/ Class VI shut off.

4.1.8. Each valve shall be provided with a bolted gland type stuffing box. Packing material shall be suitable for the process temperature. PTFE shall be used as standard packing material for bonnet temperatures below 230 degC and graphite for higher temperatures. Higher temperatures can be accepted for PTFE if the bonnet is extended. Considering fire proof, graphite packing shall be used.

4.1.9. Valve connection shall be as per piping material specification. For BBOU and ELEP, all SDVs and BDVs are of 900# RTJ.

4.2. Actuators

4.2.1. Gas Over Oil (GOL) ESD/BDV valve shall be used. GOL valve shall have external gas cylinder that shall act 'instrument air' for the operation of the valve. Dry natural gas or dry air shall be used in the external cylinder. External Gas cylinder shall be equipped with Pressure Gauge, Pressure Transmitter for detecting the low pressure in the cylinder and to trigger the re-fill the cylinder. External Cylinder shall be sized for the three (3) full valve cycles. Cylinder shall be designed as per ASME Section VIII and shall have pressure safety valve.

4.2.2. All ESD valve shall be 'Fail Close' and BDV valves shall be 'Fail Open'. Hand wheel shall be avoided for ESD/BDV valves.

4.2.3. Torque requirements for the actuator at any point in the stroke depend on the type of application. In selecting an actuator, to meet the torque requirements, all factors shall be taken into consideration to ensure that the delivered actuator meets the required torque output but is not significantly over-sized. Vendor shall be responsible for the right selection of the actuator based on the process data and 'approved' purchaser's datasheets.

4.2.4. Actuator shall be sized for the shut-off differential pressure indicated in approved purchaser's datasheets and @ minimum pneumatic supply of gas cylinder pressure.

4.2.5. The vendor shall provide the external gas cylinder for the actuation of the ESD/BDV valves. This cylinder shall be periodically filled with portable cylinder with hand pump.

Service	Minimum Safety Margin (of Max.Shut off DP) **
All ESD/BDV valves	a) SST/VSCT : 2.0 b) SRT/VRT: 1.5 c) SET/VRST:1.25 d) MAST/MST:1.1

4.2.6. Safety Margin shall be as per the Safety Margin Selection Matrix provided in the following table.

SST : Spring Start Torque

VSCT : Valve Start To Close

SRT : Spring Running Torque

VRT : Valve Running Torque

SET : Spring End Torque

VRST : Valve Re-seat Torque

MAST : Maximum Allowable Stem Torque

MST : Maximum Spring Torque.

4.3. Solenoid valves

4.3.1. Two nos. of SIL-3 solenoid valves shall be provided and must have sufficiently sized CV so as not to impede the valve stroking time. All reliability parameters like PFD values, SFF etc shall be obtained from OEM for SIL calculation / verification during detail engineering.

4.3.2. Solenoid valve shall be explosion proof. Class H insulation allowing continuous operation in an 85°C ambient temperature shall be used in solenoid valve.

4.3.3. Quick exhaust valves shall be utilised to improve valve response time.

4.4. Valve Limit Switch

4.4.1. Limit Switches shall be Mechanical Limit Switches with SPDT signals and as per NAMUR.

4.4.2. Material of valve top box shall be Die cast aluminium with Epoxy painting.

4.5. Air/Gas Filter Regulator

4.5.1. Vendor shall supply Air/Gas Filter Regulators with metal bowl and with 40 micron or better filter element, removable drain plug and an output gauge.

4.5.2. The Air filter regulator shall be of suitable range, installed and piped with actuator Pressure at the inlet of Air filter Regulator shall be 7 bar g. Inlet connection shall be 1/4" NPT(F) unless higher size is required to meet specified conditions.

5. Special Test Requirement

This specification identifies special tests to be carried out at vendor's works in addition to the normal tests for ESD/BDV Valves. Vendor to furnish the test details of special tests for approval. All testing shall be as per the approved Quality Assurance Plan (QAP).

5.1.1. Helium leak test shall be carried out for ESD/BDV on Hydro Carbon Service. Vendor shall furnish :

-Test Pressure

-Duration of the Test

-Allowable Leakage Rate

5.1.2. Radiography Test

5.1.3. Magnetic Partical Test

5.1.4. Ultrasonic Examination

6. Tag Nos.

Sr. No.	Tag No.	Application	Line Size and Rating	Shut off Press.	Body Body and Trim MOC	Shut off Class	Fail Position
1	BBOU-SDV-1001	Inlet Shutdown Valve-BBOU Gathering Manifold	24",900# RTJ	110 Barg	A 105, SS 316	TSO	FC
2	BBOU-BDV-1002	Blowdown Valve to Vent	4",900# RTJ	110 Barg	A 105, SS 316	TSO	FO
3	BBOU-SDV-1003	Outlet Shutdown Valve- Pig Launcher BBOU-A-200	24",900# RTJ	110 Barg	A 105, SS 316	TSO	FC
4	ELEP-SDV-1001	Inlet Shutdown Valve-ELEP Gathering Manifold	36",RTJ	110 Barg	A 105, SS 316	TSO	FC
5	ELEP-BDV-1002	Blowdown Valve to Vent	4",900# RTJ	110 Barg	A 105, SS 316	TSO	FO
6	ELEP-SDV-1003	Inlet Shutdown Valve-OPUGBENE Pig Receiver ELEP-A-200	24",900# RTJ	110 Barg	A 105, SS 316	TSO	FC
7	ELEP-SDV-1004	Outlet Shutdown Valve- Pig Launcher ELEP-A-300	36",RTJ	110 Barg	A 105, SS 316	TSO	FC
8	AKAB-SDV-1003	Inlet Shutdown Valve-OPUGBENE Pig Receiver AKAB-A-100	36",RTJ	110 Barg	A 105, SS 316	TSO	FC

7. Recommended Vendors

7.1. Vendor shall be selected from CLIENT's approved Vendors List.

7.2. Without prejudice to the final CLIENT vendor list at the time of instrument purchase, the following additional vendors are recommended for consideration.

- Cameron with Leeden Actuator or equivalent

7.3. If above vendor is not on CLIENT's approved vendor list, he shall go through the CLIENT's pre-qualification process before being considered.

8. Documentation

8.1. Assembly drawings shall be provided for all actuated valve assemblies.

8.2. Assembly drawings shall show manufacturers, model numbers, and settings (if applicable) for all components.

8.3. Assembly drawings shall specify tubing/piping material and sizes.

8.4. An assembly drawing shall identify ports, normally energized or normally de-energized port positions, and electrical termination of components (e.g., solenoid valves).

APPENDIX – A

Codes and Standards

Standards	Description
API 598	Valve Inspection and testing
ASME B 16.10	Face-to-face and End-to-end dimensions of valves
NAMUR	For proximity switch mounting and solenoid valve connection
ASME/ANSI B16.5	Steel pipe flanges and flange fittings
ASME/ANSI B1.20.1	Pipe Threads General purpose (inch)